Department of Electrical and Electronics Engineering
National Institute of Technology, Tiruchirappalli-15

Assessment -II EEPC14 Digital Electronics –III SEM EEE-II Year

Date: 24.09.2024 Time: 09.20 am to 10.20 am Maximum Marks: 15

Answer ALL Questions
(Questions 1 to 4: each carries one mark)

1. How many 4-bit parallel adders would be required to add two binary numbers each representing decimal numbers up through 300_{10}?
2. The output Y of a 2-bit comparator is logic 1 whenever the 2-bit input A is less than 2-bit input B. The number of combinations for which output is logic 1 is__________
3. If a 3-input NAND gate has eight input possibilities, how many of those possibilities will result in a LOW output?
4. How many numbers of adjacent squares and cell numbers will be there for the square with place value'7' in a five variable Karnaugh map?
5. A BCD code is being transmitted to a remote receiver. The bits are A3A2A1A0, with A3 as the MSB.The receiver circuitry includes a BCD error detector circuit that examines the received code to see if it is legal BCD code. Design this circuit to produce a high for an error condition. Draw the logic circuit diagram using NAND gates. **(2)**
6. Implement $Y = \overline{A + B}$ using appropriate Multiplexer. **(2)**
7. Use Karnaugh maps to simplify the following Boolean expressions and Draw logic circuit diagram Using NOR Gates: **(2)**
   $a. Y = \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + A\bar{B}\bar{C} + A\bar{B}C$
   $b. Y = \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}C\bar{D} + A\bar{B}\bar{C}\bar{D} + A\bar{B}C\bar{D}$
8. Design a hexadecimal to seven segment decoder circuit with common Anode configuration to display the hexadecimal numbers as per the details show in fig.1and Fig .2. Draw the logic circuit diagram using logic gates. **(4)**

0 000
0 0 – 0
0 0 0
0 0 0

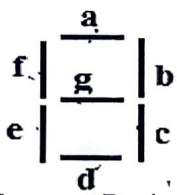

Fig.1 Segment Designation

0010
0001
0011
~~10~~
0110
0111
1011

Hexadecimal Digit Display Format

Fig.2

0	1	3	2
4	5	7	6
12	13	15	14
8	9	11	10

16	17	19	18
20	21	23	22
28	29	31	30
24	25	27	26

	Y
000	1
001	1
010	1
011	1
100	1
101	1
110	1
111	0

00111

10111

1(C+)
= 6